

Mole Relationships in Chemical Equations

Physical Science

Mole Relationships in Chemical Equations

250g rice + 250g water $\longrightarrow$ 500g boiled rice

Conservation of mass:

mass of reactant = mass of product

$$500\text{g} = 500\text{g}$$

Mole Relationships in Chemical Equations

A hydrogen gas reacts with Nitrogen gas to obtain a product ammonia.



- Is the equation balance?

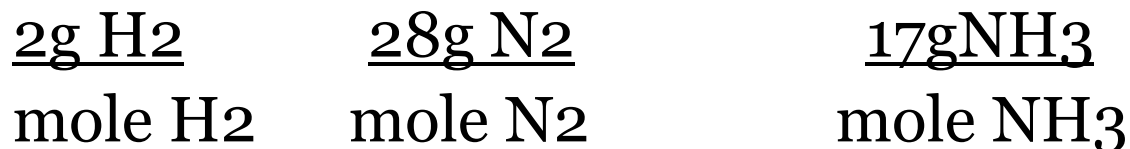


Mole Relationships in Chemical Equations

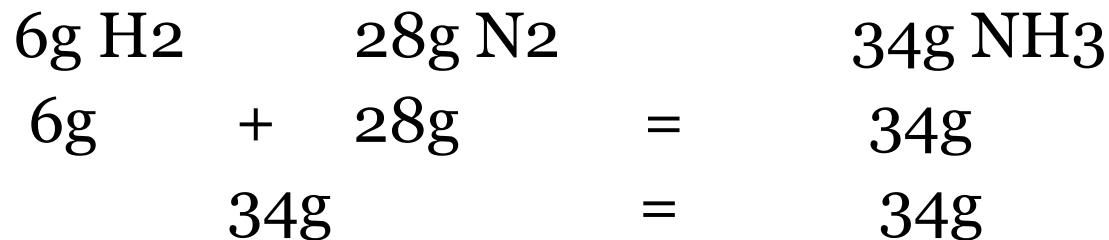


3moles H₂ + **1 mole** N₂ *forms* **2 moles** NH₃

In molar mass you get:



So, the total mass:



Seatwork

- Ag(silver) reacts with S(sulfur) to form Ag_2S (silver sulfide). How many moles of Ag_2S is produced from 2 moles of Ag?

$$\text{Ag}=108$$

$$\text{S}=32$$

Limiting Reactants

- the amount of material that limits the amount of the product that can be formed by the reaction
- After all the limiting reactant is consumed, the product will stop to form
- The excess reactant will be left in the reaction.

Example:

- Preparation of hotdog on stick with marshmallow
- Each stick should contain 2 marshmallows and 1 hotdog according to recipe.
- Suppose you have **10 sticks**, **25 marshmallows** and **9 hotdogs**, the number of hotdogs limits the number of hotdog on sticks that can be assembled according to the list of ingredients needed in the recipe. The marshmallow is the **excess reactant**.

Limiting Reactants

In chemical reactions, how do you determine the limiting reactant?

Consider the following reaction:



Solution

- Determine the amount of NH_3 that forms in this reaction when 84.06g of N_2 and 22.18g H_2 react.
- Convert to moles the amount in grams of the reactants:

$$\text{N}_2: \frac{84.06\text{g N}_2}{28.02\text{g/mole N}_2} = 3 \text{ moles}$$

$$\text{H}: \frac{22.18\text{g H}_2}{2.016\text{g/mole H}_2} = 11.002 \text{ moles}$$

Solution

- In the balanced equation:
1 mole N_2 react with 3 moles H_2 to produce 2 moles NH_3 . Therefore, 3 moles N_2 will react with 9 moles H_2 to form 6 moles NH_3 .

H_2 is excess reagent by 2.002 moles.

N_2 is the limiting reagent.

Reaction Yield

Theoretical Yield

- is the amount of the product that you can obtain in a reaction when stoichiometry is used to calculate it.
- It is the theoretical yield that forms when all limiting reagent reacts to form the desired product.

Actual Yield

➤ is the amount of product that actually formed during the reaction

❖ How do we compute the % yield?

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{Theoretical yield}} \times 100\%$$

Percent Yield

➤ is a measure of the efficiency of a reaction carried out in the laboratory

Factors that could affect % yield of a reaction:

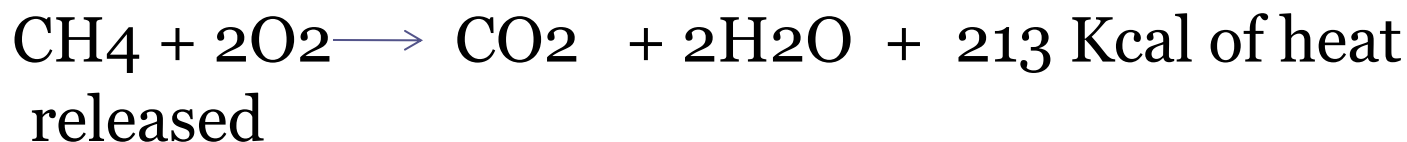
1. Incomplete reaction
2. Impure reactants
3. Competing side reactions and
4. Loss of product during filtration in transferring between containers

Energy in Chemical Reactions

- In a chemical reaction, energy is either given off or absorbed.
- When the energy of the products is lower than that of the reactants, heat energy is given off and the reaction is called **exothermic**.
- If the energy of the product is higher than the energy of the reactants, heat is absorbed and the reaction is called **endothermic**.

Examples:

- Heat is released during the process. (Exothermic Reaction)



- Heat is absorbed when hydrogen and iodine undergo chemical reaction. (Endothermic Reaction)



Evaluation

- What is the difference between endothermic reaction and exothermic reaction?
- What is theoretical yield?
- What is percent yield?